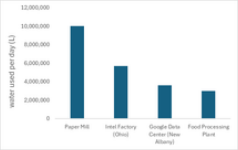
Data centers have a devastating impact on local water infrastructure.

Data centers plunder water while generating no local employment.

While data centers serve as indispensable infrastructure supporting cloud services and AI, they also constitute an industry that consumes vast quantities of water. A particularly critical issue is the cooling process required to dissipate the heat generated by servers. Many large-scale facilities employ evaporative cooling or water-cooling systems; consequently, it is not uncommon for a single facility to consume anywhere from hundreds of thousands to millions of liters of water per day. In certain regions, this level of water usage places a significant strain on local resources.



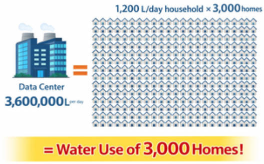
Graph
Daily Water Consumption by Major Industries

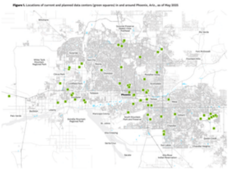
For instance, in the vicinity of New Albany, the simultaneous expansion of data centers—alongside the construction of an Intel semiconductor plant and a growing population—has triggered a rapid surge in demand for limited water resources. Google's data center in the area, for example, consumes 3.6 million liters of water daily—an amount equivalent to the water usage of an entire manufacturing plant. Furthermore, the issue extends beyond mere volume; because water demand is heavily influenced by ambient temperature, usage spikes sharply during the summer months, thereby straining peak water-supply capacity.

As this graph illustrates, data centers consume more water than typical meat processing plants. While semiconductor factories and similar facilities also utilize large quantities of water, their manufacturing lines employ a substantial workforce; in contrast, data centers do not generate local employment. In short, data centers consume vast amounts of water—much like other industrial facilities—yet they fail to deliver economic benefits to the local community.

In increasingly arid Arizona, water consumption by data centers is on the rise.

In the American Southwest, specifically in the Phoenix metropolitan area of Arizona, the rapid proliferation of data centers has recently coincided with increasing constraints on water resources. Because the region is naturally arid and relies heavily on rivers and groundwater, water is effectively treated as a finite resource. Against this backdrop, driven by the expanding demand for cloud computing and AI technologies, major corporations such as Microsoft, Google, Meta, and Amazon Web Services have established data centers in the area. This has created new pressures on local water usage.





Map of Data Center in Phenix

Cooling is essential for data centers to dissipate the heat generated by servers. Facilities that use evaporative cooling systems consume water directly. A single facility can consume tens of millions of gallons of water annually—a volume equivalent to approximately 100 to 400 million liters per year. Around 2020, data center water consumption across the entire Phoenix metropolitan area was estimated at approximately 385 million gallons (1.46 billion liters) per year. However, by around 2025, this figure is believed to have surged to approximately 900 million gallons (3.4 billion liters) per year. In other words, consumption more than doubled in just a few short years.

Data centers consume vast amounts of electricity, and the power plants that generate this electricity also require water. Since thermal and nuclear power plants require substantial quantities of water for cooling purposes, the actual "water footprint" of data centers becomes significantly larger when indirect water usage is taken into account.

Information that is inconvenient for companies —such as water usage figures—is concealed.

A key issue surrounding data center water use is the lack of transparency, especially regarding confidential agreements between technology companies and local governments or utilities. Although companies like Microsoft, Google, Meta, and Amazon Web Services publish sustainability reports, the information is usually aggregated at the national or global level. These reports lack facility-specific or contract-level detail. Often, data centers are developed under economic or non-disclosure (NDA) agreements that restrict the public availability of critical information. These agreements can restrict access to details such as site-specific water consumption, peak seasonal usage, water pricing structures, and infrastructure cost-sharing arrangements. Consequently, local communities and researchers often cannot accurately assess the environmental impact of individual facilities. This lack of transparency creates several problems.

City	Information that has been inaccessible
Goodyear, Arizona (Microsoft)	• Detailed Water Usage Breakdown • Percentage of Reclaimed Water Utilization • Cooling Methods
Mesa, Arizona (Facebook)	• Details regarding infrastructure conditions at the time of contract (specifically water and electricity) • Selected tax incentives and supply terms
New Albany, Ohio (Amazon)	• Details of Water and Power Infrastructure Contracts • Specific Criteria for Tax Incentives

Another important issue is the fragmentation of available data. Even when some information is disclosed through environmental permits or sustainability reports, it is often incomplete, inconsistent, or not directly comparable across companies. This makes it difficult to understand the cumulative impact of multiple large-scale data centers operating within the same geographic area. The core problem ultimately lies not only in the existence of confidential agreements but also in the imbalance between private contractual secrecy and public resource dependency. Water is a shared, limited resource. When its allocation is shaped by partially hidden agreements, ensuring accountability, equitable distribution, and informed public debate becomes challenging.

Reference

- Lei, Nuoa, et al. "The Water Use of Data Center Workloads: A Review and Assessment of Key Determinants." Resources, Conservation and Recycling, vol. 219, June 2025, p. 108310, https://doi.org/10.1016/j.resconrec.2025.108310.
- Lei, Nuoa, and Eric Masanet. "Climate- and Technology-Specific PUE and WUE Estimations for U.S. Data Centers Using a Hybrid Statistical and Thermodynamics-Based Approach." Resources, Conservation and Recycling, vol. 182, July 2022, p. 106323, https://doi.org/10.1016/j.resconrec.2022.106323.
- Mytton, David. "Data Centre Water Consumption." Npj Clean Water, vol. 4, no. 1, 15 Feb. 2021, www.nature.com/articles/s41545-021-00101-w, https://doi.org/10.1038/s41545-021-00101-w.